| | | | | |
|---|---|---|---|---|
| 24 | | 12.3 | P1 | for process to start,<br>e.g. correct substitution into $\frac{1}{2}ab\sin C$, |
| | | | | e.g. $0.5 \times 7 \times BC \times \sin 70 = 42$ |
| | | | P1 | (dep on P1) for process to rearrange to find $BC$,<br>e.g. $BC = \frac{84}{0.5 \times 7 \times \sin 70}$ oe (= 12.77013327) |
| | | | P1 | (dep on first P1) for process to find $AB$,<br>e.g. $AB^2 = 72 + \text{“}BC\text{”}^2 - 2 \times 7 \times \text{“}BC\text{”} \times \cos 70$ |
| | | | P1 | for correct order of operations or 150.929(30436946) |
| | | | A1 | for answer in range 12.28 – 12.3 |

**Documented scheme clarification**

1MA1_MOCK_SET_02_2H_Q24

**Original source**

scheme page 12, SHA256
73c2d2ed311e18d83d1267b625a2cb420070ca5d64f08801c018ef9003d9c2aa

**Original wording**

The source prints BC = 84/(0.5 x 7 x sin 70) while stating 12.77013327.

**Correction / interpretation**

Use BC = 42/(0.5 x 7 x sin 70), equivalently 84/(7 x sin 70), giving 12.77013327 cm. The source displayed fraction has an extra factor of 2. Then the cosine rule gives AB = 12.3 cm to 3 significant figures, as in the official final answer.

**Reason**

Rearrange 0.5 x 7 x BC x sin 70 = 42 without doubling the numerator while retaining 0.5.

**Sol authority**

51c4d6e17933058a9fe81587949167051dbebe1efcfc61bb70dea124233e14c3